INTERNAL NOTES ON
RATE MODELING
& RUN-OFF RISK

A SELF-CALIBRATING MULTIFACTOR MODEL OF THE YIELD CURVE IN ANALYSIS OF RWA

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This is a white paper on multifactor recursive update rule used in the duration adjustments in the RWA section in https://github.com/OCBliss/Basel3

Abstract. he framework models each maturity as a separate stochastic process with stochastic volatility and stochastic volatility-of-volatility while dynamically estimating cross-maturity dependence through rolling covariance matrices and Cholesky decomposition. Unlike conventional implementations that calibrate parameters and correlation structures only once before simulation, the proposed methodology recursively recalibrates model parameters and covariance matrices throughout the forecasting horizon using both historical observations and previously simulated states. This recursive update rule allows the simulated yield curve to adapt continuously as the information set evolves, providing a flexible framework for duration-sensitive applications such as interest-rate forecasting and risk-weighted asset analysis.

1. Cholesky–Banachiewicz Process

$$\Sigma(t) = L(t)L^T(t) \tag{1.1}$$

The driving Brownian motions for returns, volatilities, and vol-of-vols are correlated using the Cholesky-Banachiewicz Process, which we generalize to d-dimensional process as:

For the d-dimensional return shocks to $X_i(t)\ \forall i \in d$

$$\begin{pmatrix} W_1^X(t) \\ W_2^X(t) \\ \vdots \\ W_d^X(t) \end{pmatrix} = L_X(t) \begin{pmatrix} Z_1^X(t) \\ Z_2^X(t) \\ \vdots \\ Z_d^X(t) \end{pmatrix} \tag{1.2}$$

For the d-dimensional volatility shocks to $X_i(t)\ \forall i \in d$

$$\begin{pmatrix} W_1^\nu(t) \\ W_2^\nu(t) \\ \vdots \\ W_d^\nu(t) \end{pmatrix} = L_\nu(t) \begin{pmatrix} Z_1^\nu(t) \\ Z_2^\nu(t) \\ \vdots \\ Z_d^\nu(t) \end{pmatrix} \tag{1.3}$$

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For the d-dimensional vol-of-vol shocks to $X_i(t)\ \forall i \in d$

$$\begin{pmatrix} W_1^{\xi}(t) \\ W_2^{\xi}(t) \\ \vdots \\ W_d^{\xi}(t) \end{pmatrix} = L_{\xi}(t) \begin{pmatrix} Z_1^{xi}(t) \\ Z_2^{xi}(t) \\ \vdots \\ Z_d^{\xi}(t) \end{pmatrix} \tag{1.4}$$

where:

- L_X, L_ν, L_ξ are Cholesky decomposition matrices derived from the covariance matrices
- Z_i^X, Z_i^ν, Z_i^ξ are independent standard Brownian motions

This allows for realistic interdependence among term structure factors, their volatilities, and their vol-of-vol effects. So far, a CIR process updates the term structure factors $X_i(t) \to X_i(t+dt)$. The stochastic volatility OU process updates volatility $\nu_i(t) \to \nu_i(t+dt)$ and a stochastic vol-of-vol CIR process that updates the volatility of volatility $\xi_i(t) \to \xi_i(t+dt)$.

2. Framework for Rate Modeling

2.1. **Stochastic Differential Equations for Each Maturity Bucket.** We propose a multi-factor Heston-like CIR model to forecast raw interest rates $X_i(t)$ for each tenor $i = 1, \ldots, d$:

$$dX_i(t) = \theta(\mu - X_i(t))dt + \nu_i(t)\sqrt{X_i(t)}dW_i^X(t), \tag{2.1}$$

where:

- θ: mean reversion speed,
- μ: long-term mean,
- $\nu_i(t)$: stochastic volatility process,
- $dW_i^X(t)$: Wiener process component for X_i.

The volatility process $\nu_i(t)$ follows:

$$d\nu_i(t) = \kappa(\eta - \nu_i(t))dt + \xi_i(t)\sqrt{\nu_i(t)}dW_i^\nu(t), \tag{2.2}$$

where:

- κ: mean reversion speed,
- η: long-term mean,
- $\xi_i(t)$: stochastic volatility-of-volatility process,
- $dW_i^\nu(t)$: Wiener process, correlated with $dW_i^X(t)$ by ρ.

The vol-of-vol process $\xi_i(t)$ follows:

$$\xi_i(t) = \text{Vol}(\nu_i(t)), \tag{2.3}$$

These processes evolve with Brownian motions $W^X(t)$, $W^\nu(t)$, $W^\xi(t)$ correlated across tenors via data-driven covariance matrices.

2.2. **Dynamic Correlation via Cholesky Decomposition.** Correlation matrices $\Sigma_X(t)$, $\Sigma_\nu(t)$, $\Sigma_\xi(t)$ are computed from the historical dataset D, updated with predictions. For example:

$$\Sigma_X(t_n) = \begin{pmatrix} 1 & \rho_{12}^X(t_n) & \cdots & \rho_{1d}^X(t_n) \\ \rho_{21}^X(t_n) & 1 & \cdots & \rho_{2d}^X(t_n) \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{d1}^X(t_n) & \rho_{d2}^X(t_n) & \cdots & 1 \end{pmatrix}, \tag{2.4}$$

where $\rho_{ij}^X(t_n)$ is the correlation between X_i and X_j over a rolling window. The Cholesky decomposition is:

$$\Sigma(t_n) = L(t_n)L(t_n)^T, \tag{2.5}$$

with $L(t_n)$ recomputed at each step. Correlated Brownian motions are:

$$W^X(t) = L_X(t)Z^X(t), \quad W^\nu(t) = L_\nu(t)Z^\nu(t), \quad W^\xi(t) = L_\xi(t)Z^\xi(t), \tag{2.6}$$

where $Z^X(t)$, $Z^\nu(t)$, $Z^\xi(t)$ are independent standard normals.

2.3. **Discrete-Time Simulation with Rolling Calibration.** To forecast rates over $[t, T]$, we discretize the SDEs using Euler-Maruyama with timestep $\Delta t = (T-t)/N$, where $t_n = t+n\Delta t$ $(n = 0, \ldots, N)$. The dataset $D = \{X_i(s), \nu_i(s), \xi_i(s) \mid t_0 \le s \le t_n\}$ spans a rolling window (e.g., 252 days), updated with predictions to recalibrate parameters and correlations via maximum likelihood estimation (MLE).

For each tenor $i = 1, \ldots, d$:

$$X_i(t_{n+1}) = X_i(t_n) + \theta(\mu - X_i(t_n))\Delta t + \nu_i(t_n)\sqrt{X_i(t_n)}\Delta W_i^X(t_n), \tag{2.7}$$

$$\nu_i(t_{n+1}) = \nu_i(t_n) + \kappa(\eta - \nu_i(t_n))\Delta t + \xi_i(t_n)\sqrt{\nu_i(t_n)}\Delta W_i^\nu(t_n), \tag{2.8}$$

$$\xi_i(t_{n+1}) := \mathrm{Vol}(\nu_\mathrm{i}(\mathrm{t})) \tag{2.9}$$

with correlated increments:

(2.10)

$$\Delta W^X(t_n) = L_X(t_n)\Delta Z^X(t_n), \quad \Delta W^\nu(t_n) = L_\nu(t_n)\Delta Z^\nu(t_n), \quad \Delta W^\xi(t_n) = L_\xi(t_n)\Delta Z^\xi(t_n),$$

where $\Delta Z_i^X(t_n)$, $\Delta Z_i^\nu(t_n)$, $\Delta Z_i^\xi(t_n) \sim \mathcal{N}(0, \Delta t)$.

Forecasting Algorithm with Rolling Calibration. Initialize D with historical data from t_0 to t. For $n = 0, \ldots, N-1$:

1. Compute $\Sigma_X(t_n)$, $\Sigma_\nu(t_n)$, $\Sigma_\xi(t_n)$ from D over the rolling window, calculating pairwise correlations (e.g., $\rho_{ij}^X(t_n)$).
2. Derive $L_X(t_n)$, $L_\nu(t_n)$, $L_\xi(t_n)$ via Cholesky decomposition of $\Sigma(t_n)$.
3. Recalibrate θ, μ, κ, η, using MLE on D, updating mean reversion parameters.
4. Generate $\Delta Z^X(t_n)$, $\Delta Z^\nu(t_n)$, $\Delta Z^\xi(t_n) \sim \mathcal{N}(0, \Delta t)$.
5. Compute $\Delta W^X(t_n)$, $\Delta W^\nu(t_n)$, $\Delta W^\xi(t_n)$ using (2.10).
6. Update $\xi_i(t_{n+1})$ using (2.9).
7. Update $\nu_i(t_{n+1})$ using (2.8).
8. Update $X_i(t_{n+1})$ using (2.7).
9. Apply $X_i(t_{n+1}) = \max(X_i(t_{n+1}), 0)$, etc., for non-negativity.
10. Append predictions to D: $D \leftarrow D \cup \{X_i(t_{n+1}), \nu_i(t_{n+1}), \xi_i(t_{n+1})\}$, shifting the window (e.g., drop t_0, add t_{n+1}).

(11) Update filtration: $\mathcal{F}^1_{t_{n+1}} = \sigma(\mathcal{F}^1_{t_n} \cup \{X_i(t_{n+1}), \nu_i(t_{n+1}), \xi_i(t_{n+1})\})$, $\mathcal{F}^2_{t_{n+1}}$ from $\Sigma(t_{n+1})$.

This approach uses MLE to adapt mean reversion parameters (θ, μ, etc.) alongside correlations, ensuring forecasts reflect current dynamics.

2.3.1. *The Update Order.* A crucial modeling point is that you cannot update the new shock to ξ before knowing the updated $\mathcal{F}^2_{t+s}$-measurable covariance matrices. Concretely:

(1) At time t, you know $\Sigma_\xi(t)$ from $\mathcal{F}^2_t$.
(2) Moving from t to $t+s$, *first* update $\Sigma_\xi(t+s)$ (thus extending $\mathcal{F}^2_t$ to $\mathcal{F}^2_{t+s}$).
(3) Then you draw the new increment for ξ based on the updated matrix $\Sigma_\xi(t+s)$. Symbolically:

$$\xi\big(t+s\big) \;=\; \xi(t) \;+\; \text{drift} \times s \;+\; \Sigma_\xi\big(t+s\big)\,\Delta W^\xi.$$

(4) Once ξ is updated, you proceed to update ν (using $\Sigma_\nu(t+s)$) and then X (using $\Sigma_X(t+s)$), always respecting $\mathcal{F}^\xi \subseteq \mathcal{F}^\nu \subseteq \mathcal{F}^X \subseteq \mathcal{F}^1$.

This ensures the covariance/volatility matrices are in place *before* the shocks to ξ, ν, X are applied at each step.

2.3.2. *Merging into One Big Filtration.* Often, one merges $\mathcal{F}^1$ and $\mathcal{F}^2$ into a single "master" filtration:

$$\mathcal{F}_t \;=\; \sigma\Big(\mathcal{F}^1_t \;\cup\; \mathcal{F}^2_t\Big).$$

Then each process ξ, ν, X sees the updated matrix $\Sigma_\xi, \Sigma_\nu, \Sigma_X$ as part of $\mathcal{F}_t$. In implementation terms, at each small time increment:

(1) Update $\mathcal{F}^2_{t+s}$ to get $\Sigma_\xi(t+s)$, $\Sigma_\nu(t+s)$, $\Sigma_X(t+s)$.
(2) Use $\Sigma_\xi(t+s)$ to drive the new shock for ξ. That places $\xi(t+s)$ in $\mathcal{F}^\xi_{t+s} \subseteq \mathcal{F}^1_{t+s}$.
(3) Use $\Sigma_\nu(t+s)$ *and* the updated $\xi(t+s)$ to update $\nu(t+s)$.
(4) Finally, use $\Sigma_X(t+s)$ *and* the updated $\nu(t+s)$ to update $X(t+s)$.

Thus, $\mathcal{F}^2$ "feeds into" the stepwise chain for ξ, ν, X.

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